

Hirna Derege

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EDUCATION

Seattle University

Bachelors of Science in Computer Science, Minor: Mathematics

Cumulative GPA: 3.2

Seattle, WA

December 2026

Relevant Coursework: Programming and Problem Solving • Data Structures • Fundamentals of Databases • Computing Systems • Object-Oriented Development • Languages and Computation • Algorithms • Technical Communication • Web Development • Big Data Analytics • Machine Learning

TECHNICAL SKILLS

Languages: C++, CUDA, C#, Python, HTML, CSS, R, JavaScript, Java

Developer Tools: VSCode, IntelliJ Idea, Eclipse, Git, JetBrains Rider, JetBrains Clion

Technologies/Frameworks: Linux, GitHub, .NET

Skills: Organized, Leader-type, Teamwork, Problem Solving, Game Development, CodePath Technical Interview Prep

PROJECTS

BiliMetrix (Senior Capstone Project)

Sept. 2025 - Jun. 2026

- Calculates bilirubin levels in blood from a snapshot of a test strip to diagnose and prevent kernicterus
- Created and optimized detectors to improve readings, updated UI in Summary Page
- Mainly written in C#, learned how to work with MAUI, Xamarin, XML, and about mobile apps

Thread-Safe GPU Allocator (Research Project)

Jun-Aug. 2025

- Enabled safe concurrent memory allocation across 4,000+ GPU threads through lock-free synchronization
- Sole researcher and developer on faculty-lead research project
- Achieved millions of memory operations per second while preventing data corruption
- Learned lock-free programming, atomic operations, used C++, CUDA, concurrent debugging tools

Live Visualizer

Aug. 2026

- Built a browser-based trace visualizer replaying 4,000,000+ real execution events, showing live memory allocation, thread contention, and retry patterns across 128, 512, and 4,096 thread configurations
- Designed custom Canvas/JS rendering engine with typed arrays and requestAnimationFrame for smooth 60fps playback of execution traces

AntHill

Aug. 2026

- Built a cross-platform iOS/Android navigation app in React Native and TypeScript
- Sole developer on personal project
- Implemented live GPS tracking and auto-advancing turn-by-turn instructions using expo-location, calculating real-world distance between coordinates via the Haversine formula to detect step completion
- Built a custom SVG-based elevation profile chart from scratch (no charting library) to visualize climb/descent along each route option

WORK EXPERIENCE

Coding with Kids

Seattle, WA

Instructor

Jun. - Aug. 2024, 2026

- Instructed over 75 students in programming fundamentals with high student satisfaction
- Mentored beginning coders who successfully completed their first programming projects
- Fostered increased interest in STEM through engaging, hands-on coding instruction
- Classes include-Minecraft modding, Scratch, intro to web development, and intro to programming

ADDITIONAL INFORMATION

GWC - Girls Who Code, member of Seattle University chapter

NSBE - National Society of Black Engineers, member of Seattle University chapter

SWE - Society of Women Engineers, member of Seattle University chapter

YEOC - Young Executives of Color, Alum

CodePath - Technical Interview Prep Course, Alum

Honors: 2019 & 2020 Dean's List, Edmonds CC HonorRoll; 2026 Dean's List, Seattle University